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NPS/001/020 – Technical Specification for Stay Ground Anchors for Overhead Lines

1. Purpose

The purpose of this document is to outline the technical requirements for stay anchor systems for overhead line distribution networks of CE Electric UK.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
NPS/001/020	1.0	Technical specification for stay ground anchors for overhead lines

2. Scope

This document applies to products for both the AB Chance Anchor Drive System and the Platipus Utility System. Both systems are currently in use within CE Electric UK.

This document includes details for specified materials and sizes to national and industrial standards.

3. Technical Requirements

3.1 General

Ground anchor systems shall generally satisfy the requirements of ENA TS 43-91: Issue 5 – 2004.

Within CE Electric UK there is a requirement for two systems for the application of stay anchor systems: -

- a) AB Chance Anchor Drive System
- b) Platipus Utility System.

Products from alternative suppliers that can be applied with the use of existing CE Electric UK plant and machinery are acceptable. Specific requirements are detailed in Appendix 1.

3.2 AB Chance Anchor Drive System

Products are required for the Mid-strength Series Pisa Anchor Augured System. They include 200mm, 250mm and 300mm single square helix with a 75mm pitch and a 6000 foot-pound typical working torque.

They are used in conjunction with 25mm diameter rods, 1065mm and 2135mm in length. The rods should be capable of being extended by means of a coupler and have a UTS of 133.4KN.

The stay is terminated onto rods with a turnbuckle barrel that comes complete with a thimble and lock nut.

Alternative offers may be considered providing compliance with the above requirements can be proven.

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3.2.1 Materials

Rods will be manufactured from Mild Steel Grade 42A in compliance with BS 7886.

Tubes will be manufactured from Heavy Gauge Steel tube in accordance with BS EN 10255.

3.2.2 Finish

All steel parts shall be galvanised to BS EN 1461. External threads are to be brush galvanised with internal threading carried out after galvanising. Threads shall be coarse pitch class 7H/8g to BS 3643 and supplied with an adequate coating of oil.

3.3 Platipus Utility System

The Platipus percussion driven B6 anchor shall be supplied with a 2M x 10mm plastic impregnated tendon. The means of adjustment is by a M20 x 230mm tubular bodied turnbuckle with a threaded oval eye each end complete with two lock nuts. The stay attachment is by means of a thimble manufactured to ENA TS 43-91 Fig. 8.1.

The anchor assembly shall have the ability to attain a minimum proof load value of 30KN with the ability to achieve a theoretical minimum failing load of 65KN.

Alternative offers may be considered providing compliance with the above requirements can be proven.

3.3.1 Materials

The anchor shall be manufactured from Cast Spheroidal Graphite Iron in accordance with BS EN 1563.

The tendon shall be manufactured from 7 stranded, herringbone construction, steel wires in accordance with BS 302 and impregnated with black UV stable polypropylene that also forms a 1.5mm outer coating.

The turnbuckle shall be manufactured in accordance with BS EN 10255.

The stay thimble shall be manufactured in accordance with BS EN 13411.

3.3.2 Finish

All steel parts shall be galvanised to BS EN 1461 after machining.

3.4 Installation

AB Chance Anchor Drive and Platipus Systems shall be installed in accordance with NSP/004/104 - Guidance on the type and installation requirements for stays.

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4. References

4.1 External Documentation

Reference	Title
ENA TS 43-91: Issue 5 - 2004	Stay Strands and Stay Fittings for Overhead Lines
BS 7668: 1994	Specification for Weldable Structural Steel
BS EN 10255: 2004	Specification for Screwed and Socketed Steel Tubes
BS 3643: 1981	ISO Metric Screw Threads
BS EN 1563: 1997	Founding, Spheroidal Graphite Cast Iron
BS 302: 1987	Stranded Steel Wire Ropes
BS EN 1461: 1999	Welding and Allied Processes

4.2 Internal documentation

Reference	Title
NSP/004/104	Guidance on the type and installation requirements for stays

5. Definitions

Term	Definition
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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
S Salkeld	Standards Engineer	S Salkeld	23/09/08

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
G Bolden	Purchasing Manager	G Bolden	26/09/08

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

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G Hammel	Overhead Standards Manager	G Hammel	30/09/08

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	03/10/08

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C Holdsworth	Standards and Assessments Manager	C Holdsworth	26/09/08

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C Morgan	Head of Procurement	C Morgan	26/09/08

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Appendix 1 – Schedule of Items

AB Chance Anchor Drive System

Commodity Code	Description
224881	10" Single Helix Assy 1" UNC Female
224899	12" Single Helix Assy 1" UNC Female
217448	Product Code Special 03 Barrel Turnbuckle c/w Eye, Thimble & Locknut
226175	1" x 7ft Rod c/w Locknut Threaded to YE spec 1" UNC Male
222414	Coupling 18mm x 26mm Rod 1" UNC Female
226183	Rod 1050mm X1" C/W Locknut 1" UNC Mal

Platipus Utility System

Commodity Code	Description
242611	Platipus B6 Anchor: With 2M x 10mm Plastic Impregnated Tendon Attached To An M20 Turnbuckle & ENA Type Thimble